



Cyber Security & Digital Forensics

CSE - 4207

Lecture - 05

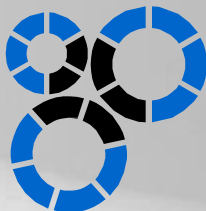
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Outline

- Polyalphabetic Ciphers
 - Autokey Cipher
 - Vernam Cipher
 - One-Time Pad
- Monoalphabetic Cipher vs Polyalphabetic Cipher



◉ Autokey Cipher

- **Autokey Cipher** is a polyalphabetic substitution cipher
- It is closely related to the Vigenere cipher but uses a different method of generating the key
- It was invented by Blaise de Vigenère in 1586
- In general, more secure than the Vigenere cipher
- But still have frequency characteristics to attack

Autokey Cipher

For Encryption:

$$C_i = (P_i + K_i) \bmod 26$$

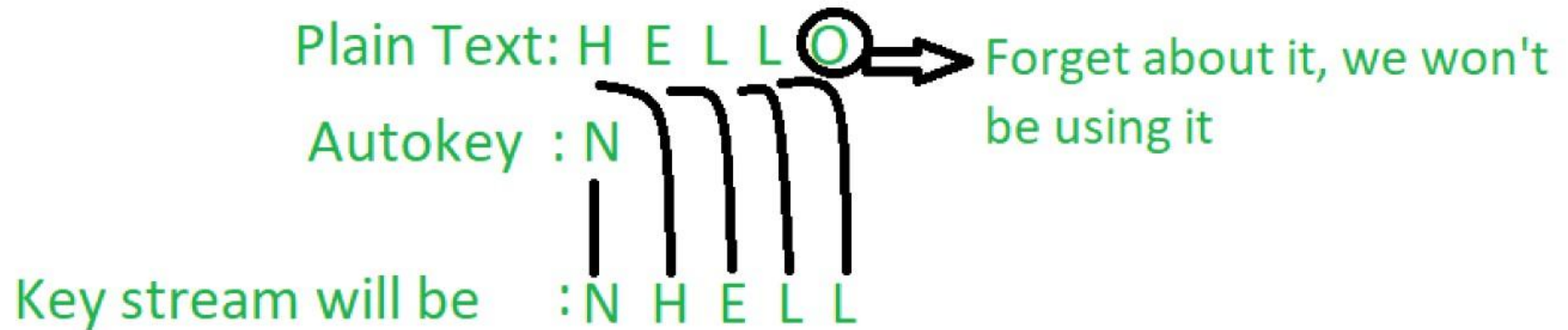
For Decryption:

$$P_i = (C_i - K_i) \bmod 26$$

Where C_i = Cipher Text Stream, P_i = Plain Text stream, K_i = Key Stream

Autokey Cipher Example

- Plaintext = "HELLO"
- Autokey = N
- Ciphertext = "ULPWZ"



Autokey Cipher Example

Given plain text is : H E L L O
Key is : N H E L L

Let's encrypt:

Plain Text(P) : H E L L O
Corresponding Number: 7 4 11 11 14
Key(K) : N H E L L
Corresponding Number: 13 7 4 11 11

Applying the formula: 20 11 15 22 25

Corresponding
Letters are : U L P W Z

Hence Ciphertext is: **ULPWZ**

Let's decrypt:

Cipher Text(C) : U L P W Z
Key(K) : N H E L L

Applying the formula: H E L L O

Hence Plaintext is: **HELLO**

0	A	9	J	18	S
1	B	10	K	19	T
2	C	11	L	20	U
3	D	12	M	21	V
4	E	13	N	22	W
5	F	14	O	23	X
6	G	15	P	24	Y
7	H	16	Q	25	Z
8	I	17	R		

Autokey Cipher Alternative Solution

		Plaintext																									
Key		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
	C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
	D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
	E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
	F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
	G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
	H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
	I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
	J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
	K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
	L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
	M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
	N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
	O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
	W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
	X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
	Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

- key: deceptivewearediscoveredsav
- plaintext: wearediscoveredsaveyourself
- ciphertext: ZICVTWQNGKZEIIGASXSTSLVWVLA

◉ Vernam Cipher

- ultimate defense is to use a key as long as the plaintext
- with no statistical relationship to it
- invented by AT&T engineer Gilbert Vernam in 1918
- originally proposed using a very long but eventually repeating key
- also known as One-Time Pad

One-Time Pad

- if a truly random key as long as the message is used, the cipher will be secure
- called a One-Time pad
- is unbreakable since ciphertext bears no statistical relationship to the plaintext
- since for **any plaintext & any ciphertext** there exists a key mapping one to other
- can only use the key **once though**
- problems in generation & safe distribution of key

◉ Vernam Cipher

- **Vernam Cipher** is a method of encrypting alphabetic text. It is simply a type of substitution cipher.
- In this mechanism we assign a number to each character of the Plain-Text, like (a = 0, b = 1, c = 2, ... z = 25).
- **Method to take key:**
In Vernam cipher algorithm, we take a key to encrypt the plain text which length should be equal to the length of the plain text.

• Vernam Cipher

- **Encryption Algorithm:**
- Assign a number to each character of the plain-text and the key according to alphabetical order.
- Add both the number (Corresponding plain-text character number and Key character number).
- Subtract the number from 26 if the added number is greater than 26, if it isn't then leave it.

○ Vernam Cipher Example

Plain-Text: RAMSWARUPK

Key: RANCHOBABA

- Now according to our encryption algorithm we assign a number to each character of our plain-text and key.

PT:	R	A	M	S	W	A	R	U	P	K
NO:	17	0	12	18	22	0	17	20	15	10
KEY:	R	A	N	C	H	O	B	A	B	A
NO:	17	0	13	2	7	14	1	0	1	0

- Add the number of Plain-Text and Key and after doing the addition and subtraction operation (if required), we will get the corresponding Cipher-Text character number.

○ Vernam Cipher Example

CT-NO: 34 0 25 20 29 14 18 20 16 10

- In this case, there are two numbers which are greater than the 26 so we have to subtract 26 from them and after applying the subtraction operation the new Cipher text character numbers are as follow:

CT-NO: 8 0 25 20 3 14 18 20 16 10

- New Cipher-Text is after getting the corresponding character from the number.

CIPHER-TEXT: I A Z U D O S U Q K

Vernam Cipher Example

Try to decrypt by reversing the algorithm!!



Monoalphabetic Cipher vs Polyalphabetic Cipher

Monoalphabetic Cipher	Polyalphabetic cipher
Once a key is chosen, each alphabetic character of plaintext is mapped onto a unique alphabetic character of a ciphertext.	Each alphabetic character of plaintext can be mapped onto “m” alphabetic characters of a ciphertext.
The relationship between a character in the plaintext and the characters in the ciphertext is one-to-one.	The relationship between a character in the plaintext and the characters in the ciphertext is one-to-many.
A stream cipher is a monoalphabetic cipher if the value of k_i does not depend on the position of the plaintext character in the plaintext stream	A stream cipher is a polyalphabetic cipher if the value of id does depend on the position of the plaintext character in the plaintext stream.
Monoalphabetic cipher includes additive, multiplicative, affine and monoalphabetic substitution cipher.	Polyalphabetic cipher includes autokey, Playfair, Vigenere, Hill, one-time pad, rotor, and Enigma cipher.



Thank You

